Lesson 26

02.10.2025

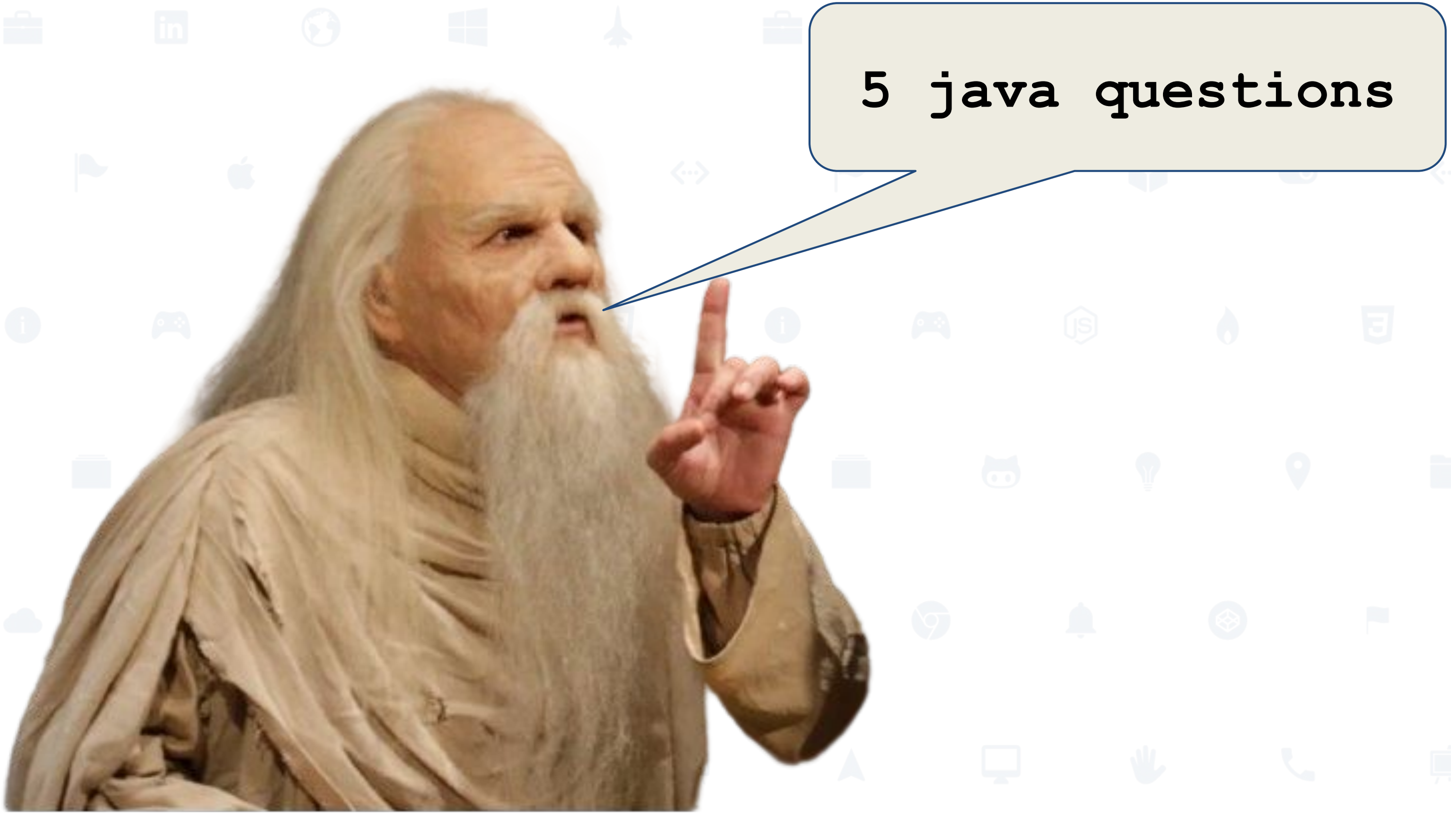
```java
public class ex1 {
    int num = 100;
    public void calc(int num)  { num = num * 10;  }
    public void printNum()     { System.out.println(num); }

    public static void main(String[] args)
    {
        ex1 obj = new ex1();
        obj.calc(2);
        obj.printNum();
    }
}
```

```java
class First {
    public First() { System.out.println("a"); }
}

class Second extends First {
    public Second() { System.out.println("b"); }
}

class Third extends Second {
    public Third() { System.out.println("c"); }
}

public class ex2 {
    public static void main(String[] args) {
        Third c = new Third();
    }
}
```

```java
public class ex3 {
    public static void main(String[] args) {
        var list = List.of(new String[]{"A", "BB", "CCC"},
                new String[]{"DD", "E"});
        list.forEach(x ->
                System.out.print(x.length));
    }
}
```

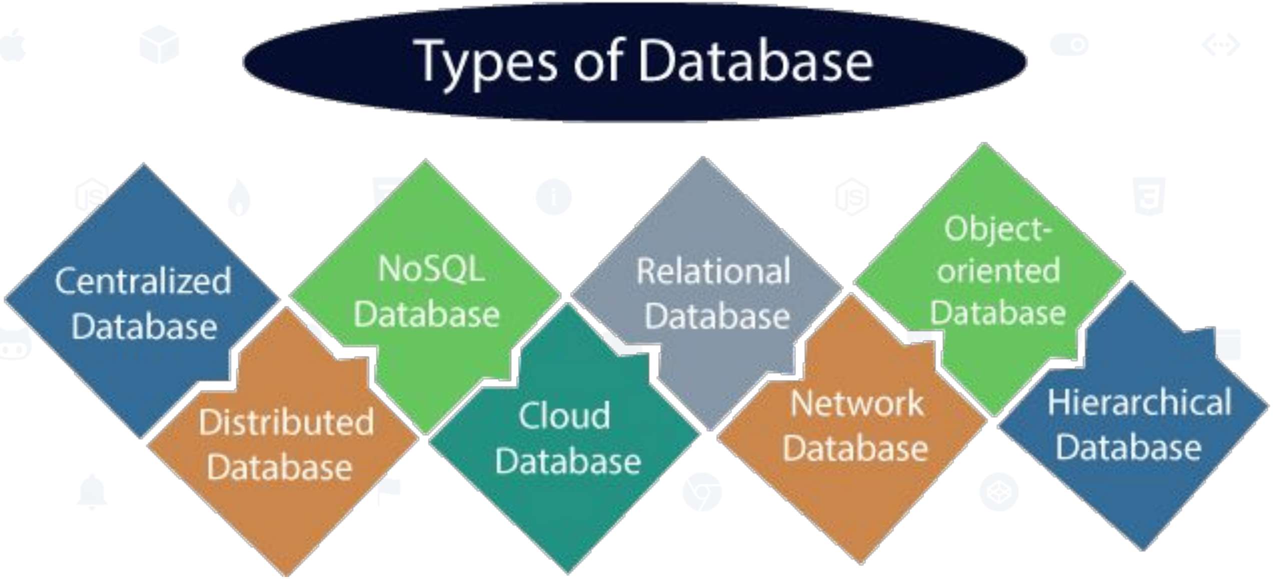
```java
public class ex4 {
    public static void main(String[] args) {
        char[][] arr = {
                {'A', 'B', 'C'},
                {'D', 'E', 'F'},
                {'G', 'H', 'I'}
        };
        for (int i = 0; i < arr.length; i++) {
            for (int j = 0; j < arr[i].length; j++) {
                System.out.print(arr[i][1]);
            }
            System.out.println();
        }
    }
}
```

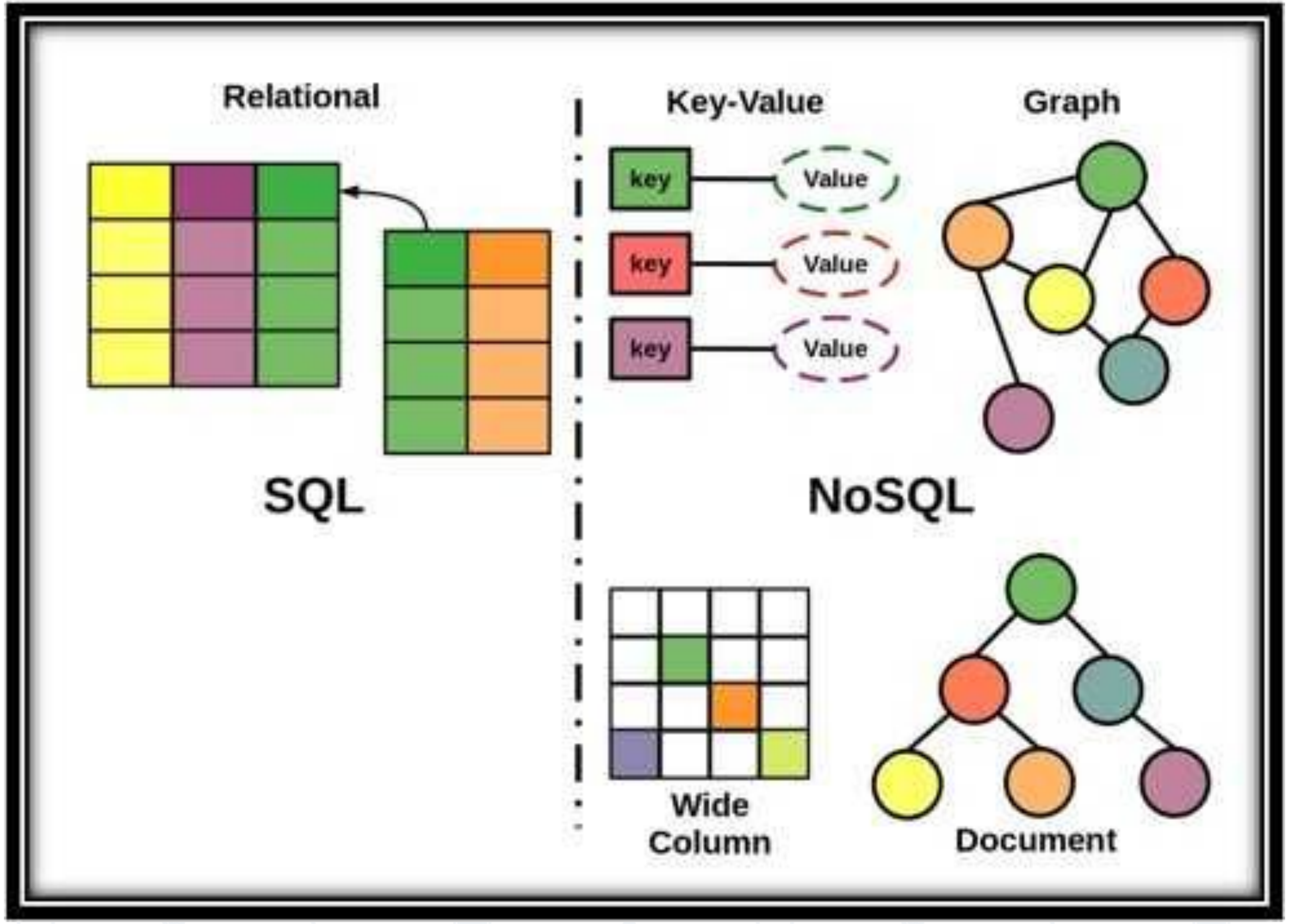
```java
class Car {
    void speed(Byte val) {System.out.println("DARK");}

    void speed(byte... vals) {System.out.println("LIGHT");}
}

public class ex5 {

    public static void main(String[] args) {
        byte b = 10;
        new Car().speed(b);
    }
}
```


5 java questions

1. Відмінності String/StringBuilder/StringBuffer
2. Interface vs Abstract Class
3. override vs overload
4. Регіони пам'яті в JVM
5. java.util.collection.

База даних - сукупність даних, що зберігаються відповідно до схеми даних, маніпулювання якими виконують відповідно до правил засобів моделювання даних.




Relational
SQL
Key-Value
key
Value
key
Value
key
Value
Graph
NoSQL
Wide
Column
Document

MariaDB

Microsoft® SQL Server®

SQLite

H2

MySQL™

ORACLE®
DATABASE

https://db-engines.com/en/ranking

425 systems in ranking, October 2025

Rank			DBMS	Database Model	Score		
Oct 2025	Sep 2025	Oct 2024			Oct 2025	Sep 2025	Oct 2024
1.	1.	1.	Oracle	Relational, Multi-model	1212.77	+42.15	-96.67
2.	2.	2.	MySQL	Relational, Multi-model	879.66	-12.11	-143.09
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model	715.05	-2.27	-87.04
4.	4.	4.	PostgreSQL	Relational, Multi-model	643.20	-13.97	-8.96
5.	5.	5.	MongoDB	Document, Multi-model	368.01	-12.49	-37.20
6.	6.	↑ 7.	Snowflake	Relational	198.65	+8.46	+58.05
7.	7.	↓ 6.	Redis	Key-value, Multi-model	142.33	-2.84	-7.30
8.	↑ 9.	↑ 14.	Databricks	Multi-model	128.80	+4.74	+43.21
9.	↓ 8.	9.	IBM Db2	Relational, Multi-model	122.37	-1.81	-0.40
10.	10.	↓ 8.	Elasticsearch	Multi-model	116.67	-1.59	-15.18
11.	↑ 12.	11.	Apache Cassandra	Wide column, Multi-model	105.16	-1.82	+7.56
12.	↓ 11.	↓ 10.	SQLite	Relational	104.56	-3.32	+2.64
13.	13.	↑ 15.	MariaDB	Relational, Multi-model	87.77	-3.69	+2.88
14.	14.	↓ 12.	Microsoft Access	Relational	80.79	-2.82	-11.36
15.	15.	↑ 17.	Amazon DynamoDB	Multi-model	75.91	-4.37	+4.06
16.	16.	16.	Microsoft Azure SQL Database	Relational, Multi-model	75.49	-3.70	+0.95
17.	17.	↑ 18.	Apache Hive	Relational	75.22	-0.88	+22.65
18.	18.	↓ 13.	Splunk	Search engine	73.87	-1.89	-17.40
19.	19.	19.	Google BigQuery	Relational	63.20	-2.80	+12.02
20.	20.	↑ 21.	Neo4j	Graph	52.51	-1.27	+10.01

Sql database structure

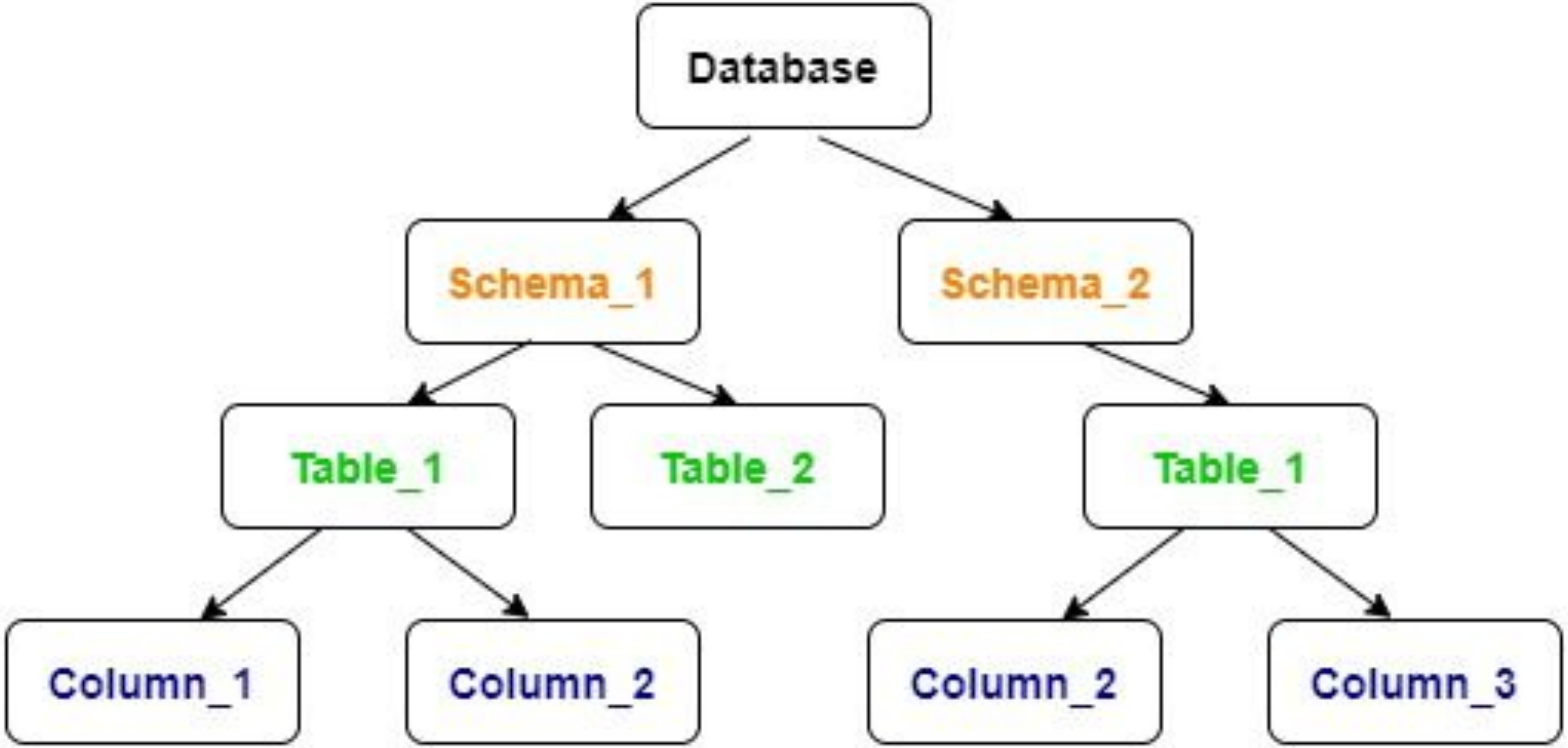
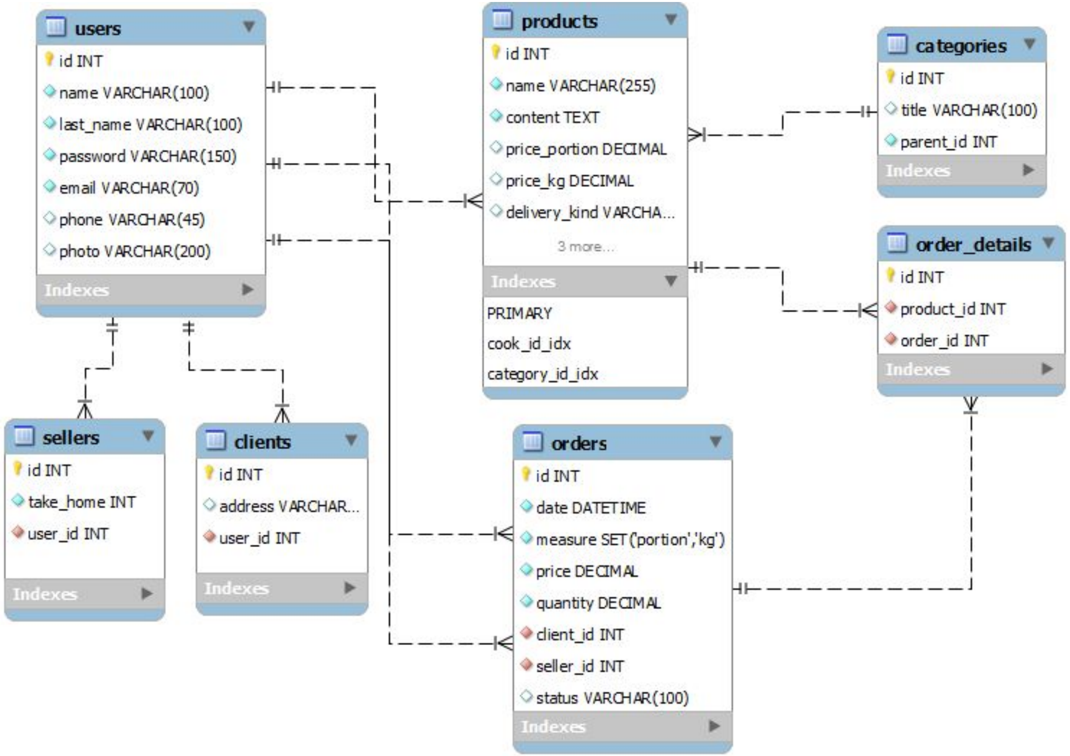
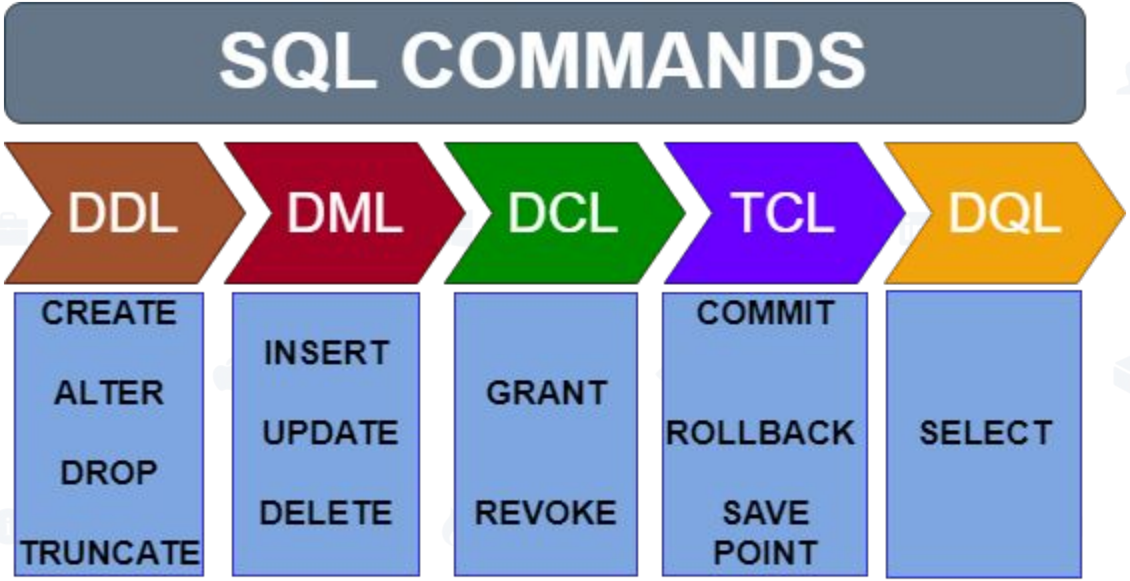


Схема бази даних





1. DDL – Data Definition Language
2. DQL – Data Query Language
3. DML – Data Manipulation Language
4. DCL – Data Control Language
5. TCL – Transaction Control Language

- **CREATE**: This command is used to create the database or its objects (like table, index, function, views, store procedure, and triggers).
- **DROP**: This command is used to delete objects from the database.
- **ALTER**: This is used to alter the structure of the database.
- **TRUNCATE**: This is used to remove all records from a table, including all spaces allocated for the records are removed.
- **COMMENT**: This is used to add comments to the data dictionary.
- **RENAME**: This is used to rename an object existing in the database.

- **SELECT**: It is used to retrieve data from the database.

- **INSERT**: It is used to insert data into a table.
- **UPDATE**: It is used to update existing data within a table.
- **DELETE**: It is used to delete records from a database table.
- **LOCK:** Table control concurrency.
- **CALL:** Call a PL/SQL or JAVA subprogram.
- **EXPLAIN PLAN:** It describes the access path to data.

```
CREATE TABLE [schema.]table
        (column datatype [DEFAULT expr][, ...]);
```

Data Type	Description
VARCHAR2(*size*)	Variable-length character data
CHAR(*size*)	Fixed-length character data
NUMBER(*p,s*)	Variable-length numeric data
DATE	Date and time values
LONG	Variable-length character data (up to 2 GB)
CLOB	Character data (up to 4 GB)
RAW and LONG RAW	Raw binary data
BLOB	Binary data (up to 4 GB)
BFILE	Binary data stored in an external file (up to 4 GB)
ROWID	A base-64 number system representing the unique address of a row in its table

```
ALTER TABLE table_name
ADD column_name datatype;
```

```
INSERT INTO table_name (column1, column2, column3, ...)
VALUES (value1, value2, value3, ...);
```

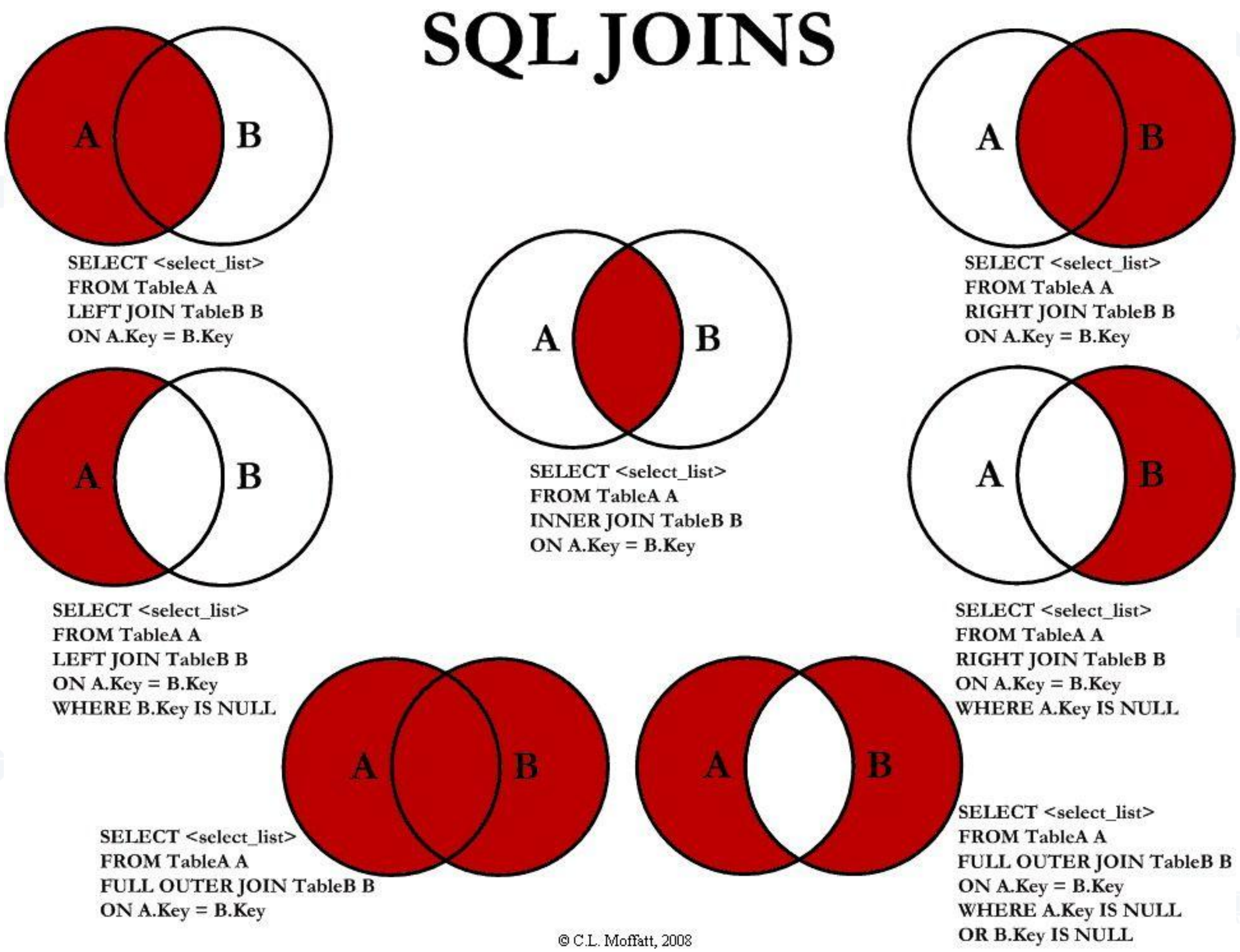
```
UPDATE table_name
SET column1 = value1, column2 = value2, ...
WHERE condition;
```

```
SELECT column1, column2, ...
FROM table_name;
```

```
SELECT COUNT(column_name)
FROM table_name
WHERE condition;
```

```
SELECT AVG(column_name)
FROM table_name
WHERE condition;
```

```
SELECT SUM(column_name)
FROM table_name
WHERE condition;
```


SQL JOINS
A
B
SELECT <select_list>
FROM TableA A
LEFT JOIN TableB B
ON A.Key = B.Key
A
B
SELECT <select_list>
FROM TableA A
INNER JOIN TableB B
ON A.Key = B.Key
A
B
SELECT <select_list>
FROM TableA A
RIGHT JOIN TableB B
ON A.Key = B.Key
A
B
SELECT <select_list>
FROM TableA A
LEFT JOIN TableB B
ON A.Key = B.Key
WHERE B.Key IS NULL
A
B
SELECT <select_list>
FROM TableA A
RIGHT JOIN TableB B
ON A.Key = B.Key
WHERE A.Key IS NULL
A
B
SELECT <select_list>
FROM TableA A
FULL OUTER JOIN TableB B
ON A.Key = B.Key
A
B
SELECT <select_list>
FROM TableA A
FULL OUTER JOIN TableB B
ON A.Key = B.Key
WHERE A.Key IS NULL
OR B.Key IS NULL
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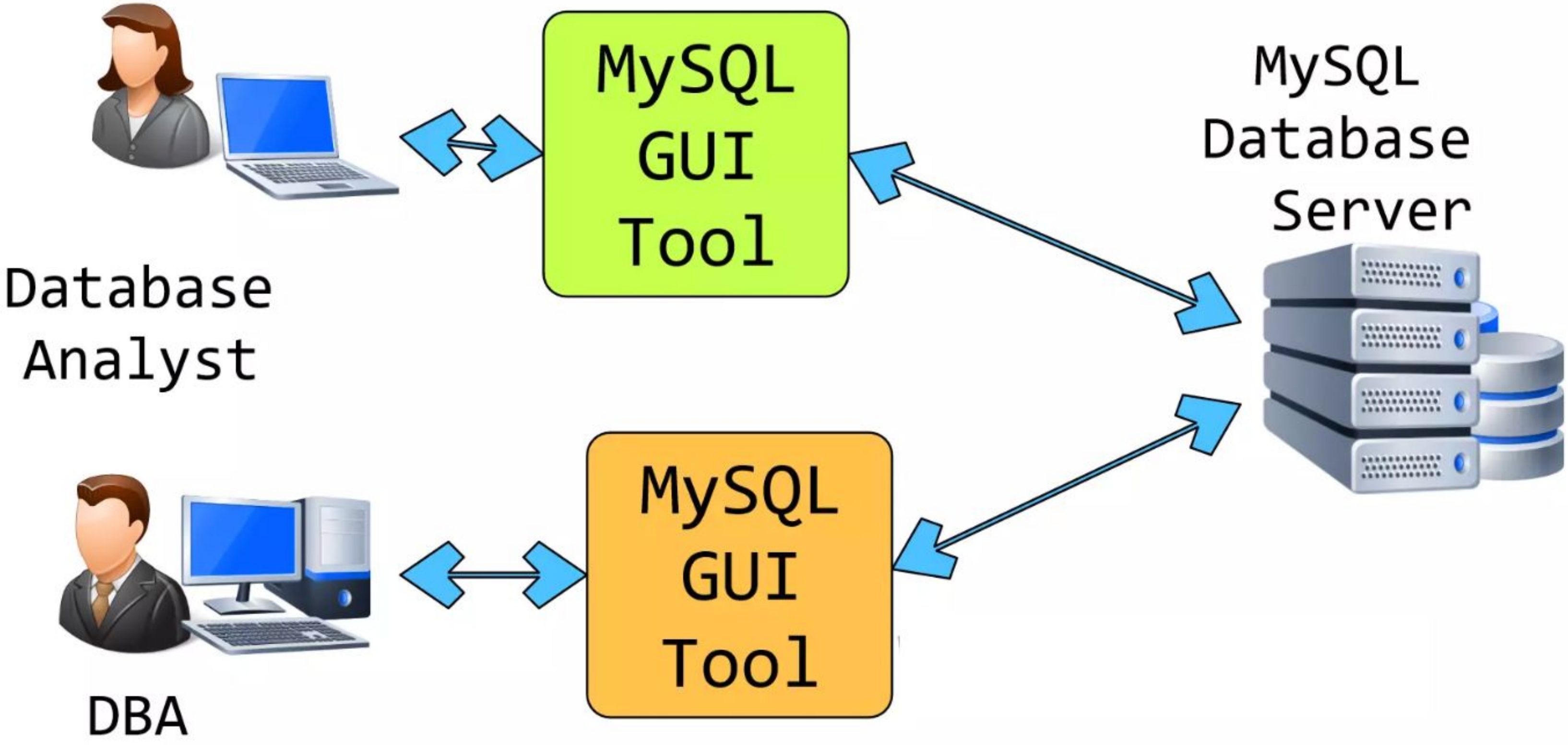
MySQL
GUI
Tool
MySQL
Database
Server
Database
Analyst
MySQL
GUI
Tool
DBA